

Supplementary Material

1. Hyperparameter Tuning and Model Selection

In addition to grid search, we tried random search for hyperparameter tuning. While random search was faster and sometimes found decent parameter combinations, grid search was more exhaustive and consistently found better-performing models for our dataset, so we used it for the results.

2. Interpretability Analysis Using LIME

To make our models more interpretable, we used LIME (Local Interpretable Model-agnostic Explanations) to understand which features influenced individual predictions. Below are examples of LIME outputs for both SVM and MLP models.

For the SVM model, LIME showed that the most important factors for approval were having no prior defaults (0.00) and a high current balance (500.00). Years employed (1.62) and savings (515.00) had smaller positive effects, while debt (25.12) and features like housing type, marital status, gender, and occupation had little to no impact. This suggests that the SVM model made decisions based on sensible financial criteria.

LIME Output for SVM: (Figure 1)

- No Prior Defaults (0.00): Strong positive impact
- Current Balance (500.00): Strong positive impact
- Years Employed (1.62): Small positive impact
- Savings (515.00): Small positive impact
- Debt (25.12): Minimal impact
- Other features: Negligible impact

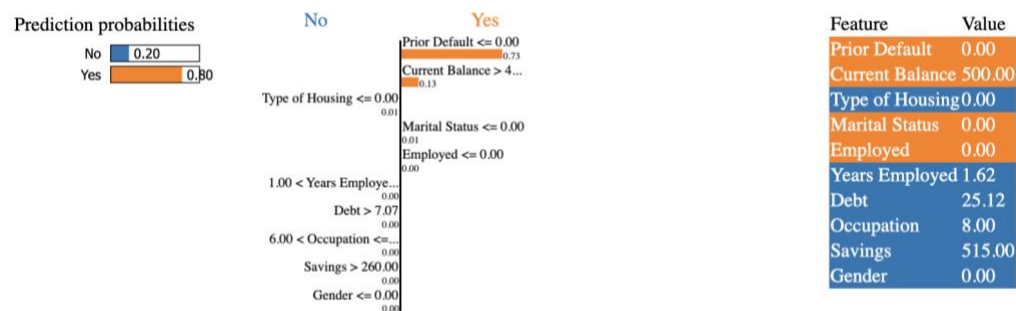


Figure 1: LIME explanation for SVM model prediction

For the MLP model, the LIME explanation was similar. The absence of prior defaults (0.00) and a high current balance (500.00) were again the most influential factors. Marital status (0.00) had a small positive effect, while other features such as gender, debt, employment status, credit score, occupation, education level, and citizenship had much less impact. This confirmed that the MLP model, like SVM, mainly relied on financial reliability when making approval decisions.

LIME Output for MLP: (Figure 2)

- No Prior Defaults (0.00): Strong positive impact
- Current Balance (500.00): Strong positive impact
- Marital Status (0.00): Small positive impact
- Other features (debt, employment, etc.): Minimal impact

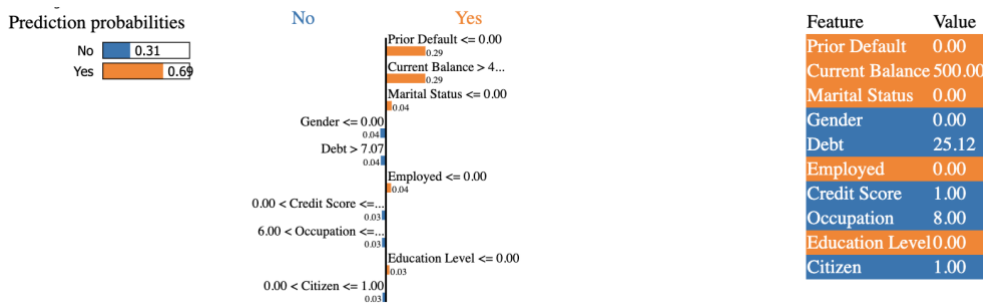


Figure 2: LIME explanation for MLP model prediction

Which Accuracy (and Metric) Matters Most?

In this project, we compared SVM and MLP models for credit card approval, mainly using metrics like accuracy, precision, recall, F1 score, and ROC-AUC. While overall accuracy is a common metric, we learned that it doesn't always reflect what actually matters in real credit approval situations.

For example, our tuned MLP model had the highest accuracy (85.50%), but the SVM model, with slightly lower accuracy (83.97%), achieved much higher recall (92.73%). This means SVM was much better at identifying all the applicants who should be approved, even if it made more mistakes by approving some risky applicants. On the other hand, MLP had higher precision (84.62%), so it was better at making sure that those it approved were actually good candidates, but it missed more people who could have been approved (lower recall: 80.00%).

Which metric is more important depends on the business context. If a bank or lender wants to grow their customer base and doesn't want to miss out on approving good applicants, recall becomes very important—so SVM would be the better choice. But if the main goal is to avoid risky approvals and potential losses, then precision is more important, and MLP might be preferred.

We also noticed that just focusing on accuracy can be misleading, especially since our dataset isn't perfectly balanced between approved and denied cases. A model could have high accuracy just by predicting the majority class, but that wouldn't help in making fair or profitable credit decisions.

This is why, in our analysis, we looked at all the metrics together and used confusion matrices and ROC curves to get a fuller picture. We also learned from the literature (IJRASET, 2022) that there's no single "best" metric—what matters most is aligning the model's strengths with the actual goals of the credit approval process.

In summary, while MLP had the highest accuracy, SVM's higher recall could be more valuable for institutions focused on growth, while MLP's higher precision is better for those focused on minimizing risk. This project showed us that choosing the right metric is as important as choosing the right model, and that decision should always be based on the specific needs of the business.